Project Documentation for

Get-DeletedTeamSites.ps1

Date: 12/12/2025

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Synopsis:
Get Deleted Teams Sites using Microsoft Graph API & Audit Data (AD Recycle Bin)

Description:
This script retrieves all deleted Microsoft 365 Groups (Teams sites) from Azure AD and correlates them with audit logs to identify who deleted them and when. The script exports results to CSV, HTML report, and sends email notifications with comprehensive logging & archiving.

This script retrieves all deleted Microsoft 365 Groups (Teams sites) from Azure AD and correlates them with audit logs to identify who deleted them and when. The script exports results to CSV, HTML report, and sends email notifications with comprehensive logging.

Current Features:
- Modular Code Design: Function-based modular design with comprehensive error handling & logging
- Audit log correlation to track who deleted Teams sites
- CSV data export with all deletion details
- HTML formatted report with summary statistics
- Email notification via Microsoft Graph API (no SMTP required)
- Automatic Timestamped Archival for Logs & Reports
- Optional: Database (SQL Server) or Data Warehouse archival of deleted sites for audit history
- Exit codes for automation (0=success, 1=failure)
- Optional: Automated database (SQL Server) or data warehouse archival of delete sites for audit history.

*** - Differentiates Teams-enabled from standard M365 Groups. ***

Links:
https://learn.microsoft.com/en-us/graph/permissions-reference
https://docs.microsoft.com/en-us/graph/api/directory-deleteditems-list
https://learn.microsoft.com/en-us/purview/audit-log-activities
https://learn.microsoft.com/en-us/graph/api/user-sendmail?view=graph-rest-1.0&tabs=http

Notes:
- Deleted groups remain in the recycle bin for 30 days and can be restored
- Audit logs are retained for 90 days

```
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  Author          : chris@brennantechnologies.com
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```

Current Process Design

Get-DeletedTeamSites.ps1

Current Process Design:

1. Modular Code Design:
- Uses a Function-based modular design for better maintainability, readability and code reuse.

2. Authentication:
- Uses an Application Registration, Certificate & 'settings.json' file for authentication.
  - ClientID, TenantID, CertificateThumbprint

3. Get Deleted Groups (Teams Sites) from MS-Graph API:
- Retrieves deleted Microsoft 365 Groups (Teams sites) from Azure AD recycle bin.
- Identifies Teams-enabled groups vs standard M365 Groups.
- Constructs SharePoint site URLs for deleted Teams sites.

4. Get Audit Logs for Deletion Tracking:
- Retrieves audit logs to identify deletion details (who, when, how).
- Correlates deleted groups with audit logs to identify who deleted them, when, and how.
- Supports configurable lookback period for audit logs (default 30 days).
- Handles scenarios where audit logs may be unavailable or incomplete.
- Retrieves detailed deletion activity types (Delete group, Remove group).
- Calculates days since deletion and days until permanent purge.
- Generates detailed objects with all relevant properties for reporting.

*** - Differentiates Teams-enabled from standard M365 Groups. ***

5. Error Checking & Handling:
- Comprehensive error handling with try/catch blocks and informative logging.

6. Logging & Reporting:
Email Report: Generates HTML report with summary statistics and detailed deletion information.
Data Export: Generate "Data File" of deleted sites in a timestamped .CSV format.
Logging: Generates detailed Log File with timestamps and severity levels.
Archives: All output files (logs, CSV exports, HTML reports) timestamped in a Logs subfolder

7. Email Notification:
- Email sending via Microsoft Graph API is used eliminating the need for SMTP server & configuration.
- Sends Email report with HTML formatted message body and CSV data file attachment.
- Configurable recipient and sender email addresses in settings.json.
- Requires Mail.Send permission in Microsoft Graph.

8. Exit Codes:
- Script returns exit codes for automation scenarios (0=success, 1=failure).

Optional Features:
- Automated database (SQL Server) or data warehouse archival of delete site results for audit history.

```
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Standard Operation Procedure

Get-DeletedTeamSites.ps1

Standard Operation Procedure:

Requirements:

- Application Registration
- Microsoft.Graph.Authentication module
- Compatible with PowerShell 5.1 and PowerShell 7+

Authentication:

- Uses certificate-based authentication via app registration
- Credentials stored in Settings.json file (ClientID, TenantID, CertificateThumbprint)

Application Registration Permissions:

```
=== REQUIRED APP REGISTRATION PERMISSIONS ===

| Permission Scope      | Type     | Required For           | Reason                                        |
|-----------------------|----------|------------------------|-----------------------------------------------|
| Directory.Read.All    | AppReg   | Core functionality     | Read deleted items from Azure AD directory    |
| Group.Read.All        | AppReg   | Core functionality     | Read Microsoft 365 Group properties           |
| Sites.Read.All        | AppReg   | -IncludeSiteInfo       | Access SharePoint site information            |
| AuditLog.Read.All     | AppReg   | -IncludeAuditData      | Read audit logs for DeletedBy, DeledDate      |
| Mail.Send             | AppReg   | Email notifications    | Send email reports (if implementing)          |
```

Settings.json File:

```
{
  "AppRegName": "M365
  "TenantId": "c603bc",
  "ClientId": "7f5ffe",
  "CertificateThumbpr537BC70380FD85554",
  "Organization": "br com",
  "SharePointAdminUrl s-admin.sharepoint.com",
  "ClientSecretID": "767465",
  "ClientSecretValue" -edQaIXOicbrT"
}
```

Required PowerShell Modules:

Required Modules: Microsoft.Graph.Authentication

PowerShell Version: Compatible with PowerShell 5.1 and PowerShell 7+

Azure Function or Scheduled Task for Automation:

- to be determined

Running Script with Parameters:

Quick Report (Minimal Output)

```
.\Get-DeletedTeamsSites.ps1
Output: Basic list of deleted Teams with core properties (DisplayName, GroupId, Mail,
DeletedDateTime, etc.)
```

Detailed Report

```
.\Get-DeletedTeamsSites.ps1 -IncludeDetails
Output: Extended properties including security settings, provisioning options, on-premises sync
status
```

Export to specific location

```
.\Get-DeletedTeamsSites.ps1 -ExportToCsv -CsvPath "C:\Reports\DeletedTeams_2025-12.csv"
```

Full export with all data

```
.\Get-DeletedTeamsSites.ps1 -IncludeDetails -IncludeSiteInfo -IncludeAuditData -ExportToCsv -
CsvPath "C:\Reports\Monthly_DeletedTeams.csv"
```

```
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Automation Workflow for:

Get-DeletedTeamsSites.ps1

Initialize & Run the Script:

Initialize Script

Requirements:
Required Modules:
- Microsoft.Graph.Authentication

PowerShell Version:
- PowerShell 5.1 and PowerShell 7+

Application Registration:
- Directory.Read.All
- Group.Read.All
- AuditLog.Read.All
- mail.send

Start → Initialize Script

Required Modules:

```
#requires -Modules Microsoft.Graph.Authentication
Import-Module Microsoft.Graph.Authentication -ErrorAction Stop
```

Required Application Registration:

```
{
  "AppRegName": "M365-PowerShell-Automation",
  "TenantId": "<tenant-id>",
  "ClientId": "<clinet-id>",
  "CertificateThumbprint": "",
  "ClientSecretID": "",
  "ClientSecretValue": ""
}
```

Script Logic:

Connect MS-Graph API

Connect-MSGraph

Required MS-Graph API Permissions:

```
$requiredScopes = @("Directory.Read.All", "Group.Read.All", "AuditLog.Read.All", "Mail.Send")
onnect-MgGraph -Scopes $requiredScopes -NoWelcome -ErrorAction Stop
```

Get Teams Data

Identifies Teams-enabled groups vs standard M365 Groups.

Get-DeletedTeamsGroups

MS-Graph Rest API Call:

```
### Build the Graph API URI for deleted groups
$uri = "https://graph.microsoft.com/v1.0/directory/deletedItems/microsoft.graph.group"
$response = Invoke-MgGraphRequest -Method GET -Uri $uri -ErrorAction Stop
```

Get-AuditData
(for "DeletedBy" & "DeletedDate")

Get Audit Log for Deleted By & Deleted Date:

```
### Search for group deletion events
$auditRecords = Search-UnifiedAuditLog -StartDate $startDate -EndDate $endDate `
                -Operations "Delete group" -ResultSize 5000 -ErrorAction Stop
```

Correlate Deleted Groups with Audit Logs

foreach ($team in $deletedTeams)

Loop through All Deleted Teams Sites

```
foreach ($team in $deletedTeams) {
        if ($IncludeSiteInfo -and $team.id) {
                ($IncludeAuditData -and $auditCache.ContainsKey($team.id)) {
                        $auditData = $auditCache[$team.id]
                }
        }
}
```

$results += $teamInfo

```
$teamInfo = [PSCustomObject]@{
                DisplayName                    = $team.displayName
                GroupId                        = $team.id
                Mail                           = $team.mail
                MailNickname                   = $team.mailNickname
                DeletedDateTime                = $team.deletedDateTime
                Description                    = $team.description
                Visibility                     = $team.visibility
                CreatedDateTime                = $team.createdDateTime
                RenewedDateTime                = $team.renewedDateTime
                ExpirationDateTime             = $team.expirationDateTime
                GroupTypes                     = ($team.groupTypes -join ', ')
                MembershipRule                 = $team.membershipRule
                MembershipRuleProcessingState  = $team.membershipRuleProcessingState
                PreferredDataLocation          = $team.preferredDataLocation
                PreferredLanguage              = $team.preferredLanguage
                ProxyAddresses                 = ($team.proxyAddresses -join '; ')
                SecurityEnabled                = $team.securityEnabled ?? $false #default
                MailEnabled                    = $team.mailEnabled ?? $false #default
                IsAssignableToRole             = $team.isAssignableToRole ?? $false #default
                Theme                          = $team.theme
                OnPremisesSyncEnabled          = $team.onPremisesSyncEnabled
                OnPremisesLastSyncDateTime     = $team.onPremisesLastSyncDateTime
                OnPremisesSecurityIdentifier   = $team.onPremisesSecurityIdentifier
                Classification                 = $team.classification
        }
```

Export Reports:

Export CSVData

Build Reports

Export-HTMLReport

Deleted Teams Data

HTML Email Report

Send Email Reports

Send-Email

Send Email with Graph API

```
### Send email via Graph API
Write-Log "Sending email report to: $($EmailTo -join ', ')" -Level Info
$uri = "https://graph.microsoft.com/v1.0/users/$EmailFrom/sendMail
Invoke-MgGraphRequest -Method POST -Uri $uri -Body $emailBody -ContentType "application/json"
```

Exit Codes & Disconnect
(Good for Automation)

Disconnect-MSGraph

Disconnect MS-Grap API

```
### Support for Automaton and Azure Functions, etc.
Disconnect-MgGraph
```

Return Exit Codes

Return Exit Codes

```
### Script returns exit codes for automation scenarios
exitcode {0=success, 1=failure}
```

End

```
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Get-DeletedTeamsSites.ps1

Summary Report - HTML Email

Deleted Teams Sites Report

Generated: 2025-12-12 14:25:55

Summary

- **Total Deleted M365 Groups:** 2
- **Teams-enabled:** 2
- **Groups only:** 0
- **With known deletion user:** 2

EXAMPLE ONLY

Deleted Teams Sites

Display Name	Deleted By	Deleted By UPN	Deleted Date	Teams Enabled	Visibility
TemplateTest1	Microsoft Teams Services	m365_test2_dev@k onmicrosoft.com	12/04/2025 17:21:07	Yes	Private
TemplateTest2	Microsoft Teams Services	m365_test2_dev@k onmicrosoft.com	12/04/2025 17:21:19	Yes	Private

Detailed information is available in the attached CSV file.

Log file: C:\Scripts\CBrennan\Get-DeletedTeamsSites\DeletedTeamsSites_20251212_142554.log

```
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Get-DeletedTeamsSites.ps1

Detailed Report

(Data .CSV Export File)

DisplayName	Description	SiteName	SiteDisplayName	DeletedBy	DeletedDateTime	Mail	SiteCreatedDateTime	SiteLastModified
M					2025-12-01T15:30:00Z		2024-09-15T09:05:32Z	2025-11-28T14:22:10Z
P					2025-11-28T10:15:00Z		2024-06-10T14:35:18Z	2025-11-20T16:45:33Z
C					2025-12-05T08:45:00Z		2024-07-01T08:10:45Z	2025-09-25T11:30:22Z
H					2025-11-20T16:00:00Z		2025-01-05T10:08:22Z	2025-11-10T09:15:44Z
Fi					2025-12-10T11:20:00Z		2024-03-20T13:12:55Z	2025-11-28T13:40:18Z
IT					2025-12-08T14:25:00Z		2024-10-01T11:15:20Z	2025-12-02T10:30:45Z
S					2025-11-15T09:00:00Z		2025-02-01T08:45:12Z	2025-11-10T16:20:30Z
P					2025-12-03T12:15:00Z		2024-08-12T10:20:18Z	2025-12-01T09:45:22Z
L					2025-11-25T13:45:00Z		2024-04-15T09:50:33Z	2025-11-20T15:10:55Z
D					2025-12-11T16:30:00Z		2025-01-15T14:18:25Z	2025-12-08T11:22:40Z
R					2025-12-02T10:00:00Z		2024-05-20T12:25:14Z	2025-11-28T14:35:20Z
S					m 2025-11-18T15:20:00Z		2024-01-10T08:30:40Z	2025-11-15T10:50:15Z
V					n 2025-12-07T11:00:00Z		2024-11-05T13:55:22Z	2025-12-05T09:40:10Z
E					2025-11-22T08:30:00Z		2025-03-01T09:20:35Z	2025-11-18T13:15:48Z
D					2025-12-09T14:00:00Z		2024-07-20T11:50:42Z	2025-12-06T16:25:33Z
M					n 2025-12-06T09:45:00Z		2024-09-01T10:30:15Z	2025-12-03T12:18:50Z
B					2025-11-30T16:15:00Z		2025-04-10T13:28:20Z	2025-11-28T10:42:15Z
Q					2025-12-04T13:30:00Z		2024-10-15T14:35:48Z	2025-12-01T15:20:32Z
R					2025-11-27T10:20:00Z		2025-02-20T09:55:30Z	2025-11-25T14:10:25Z
P					2025-12-12T15:00:00Z		2025-05-15T11:32:18Z	2025-12-10T16:55:42Z

```
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Get-DeletedTeamsSites.ps1

Log File

Log File Format Example:

```
[2025-12-13 14:30:15] [Info] Connecting to Microsoft Graph...
[2025-12-13 14:30:18] [Success] Successfully connected to Microsoft Graph
[2025-12-13 14:30:19] [Verbose] API URI: https://graph.microsoft.com/v1.0/directory/deletedItems/microsoft.graph.grou
[2025-12-13 14:30:22] [Success] Found 5 deleted Team(s)
[2025-12-13 14:30:25] [Success] Results exported to: C:\Reports\DeletedTeams.csv
```

Usage Examples:

```
# Use default log file
.\Get-DeletedTeamsSites.ps1

# Specify custom log path
.\Get-DeletedTeamsSites.ps1 -LogPath "C:\Logs\TeamsDeletion_$(Get-Date -Format 'yyyyMMdd').log"

# Full command with logging
.\Get-DeletedTeamsSites.ps1 -IncludeDetails -IncludeAuditData -ExportToCsv -LogPath "C:\Logs\DeletedTeams.log"
```

```
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